

Today

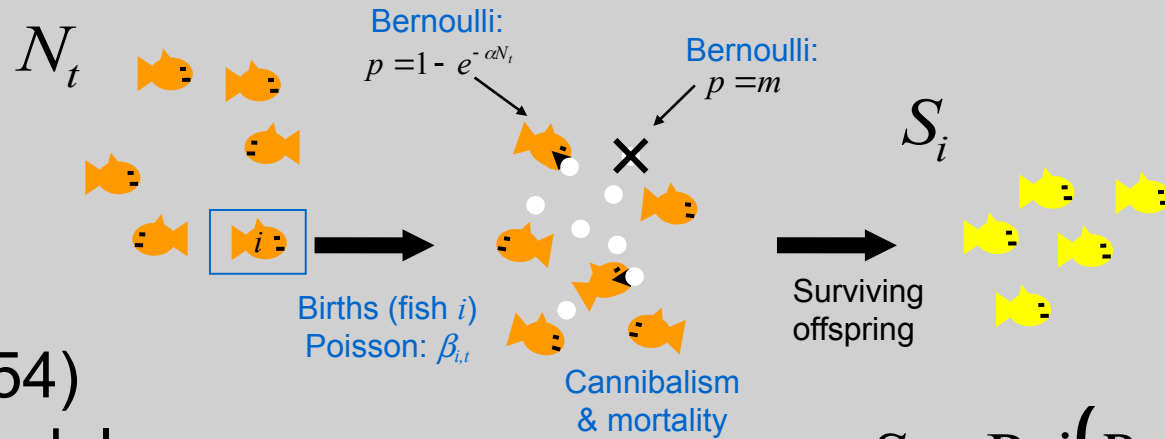
- From stochastic individuals to difference equation models
 - discrete time
- Sources of variation and stochasticity
- Combining distributions
- Time to extinction
- Dynamical systems theory
 - cycles and chaos
- Emergence of discrete "deterministic skeleton plus stochasticity" algorithm

Sources of variation

- In dynamic processes:
- **Internally-driven** variation
 - biological processes
 - variation in biological mean and stochasticity
- **Externally-driven** variation
 - “forcing” processes, environment
 - variation in environmental mean and stochasticity
- External vs internal is **scale** and **context** dependent

Sources of stochasticity

- In population dynamics:
- Demographic stochasticity
 - random birth & death events: within-individual scale
- Demographic heterogeneity
 - birth/death rates vary between individuals
- Sex (e.g. of demographic heterogeneity)
 - random event: male or female offspring?
- Environmental stochasticity
 - random birth/death events/rates: population scale

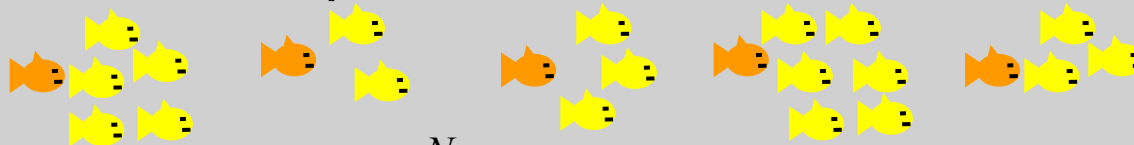


Ricker (1954)
Fisheries model

$$S_i \sim \text{Poi}(R e^{-\alpha N_t})$$

Sum up survivors over all adults

$$R = \beta(1 - m)$$

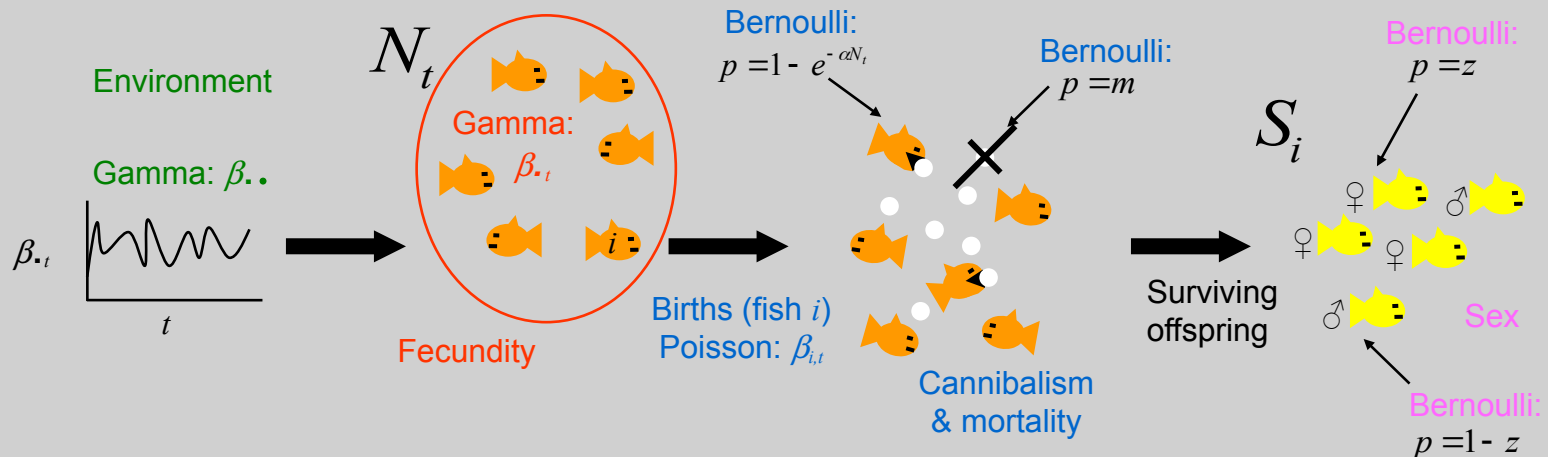


$$N_{t+1} = \sum_i^{N_t} S_i \sim \text{Poi}(N_t R e^{-\alpha N_t})$$

Poisson Ricker

(sum of Poissons is Poisson)

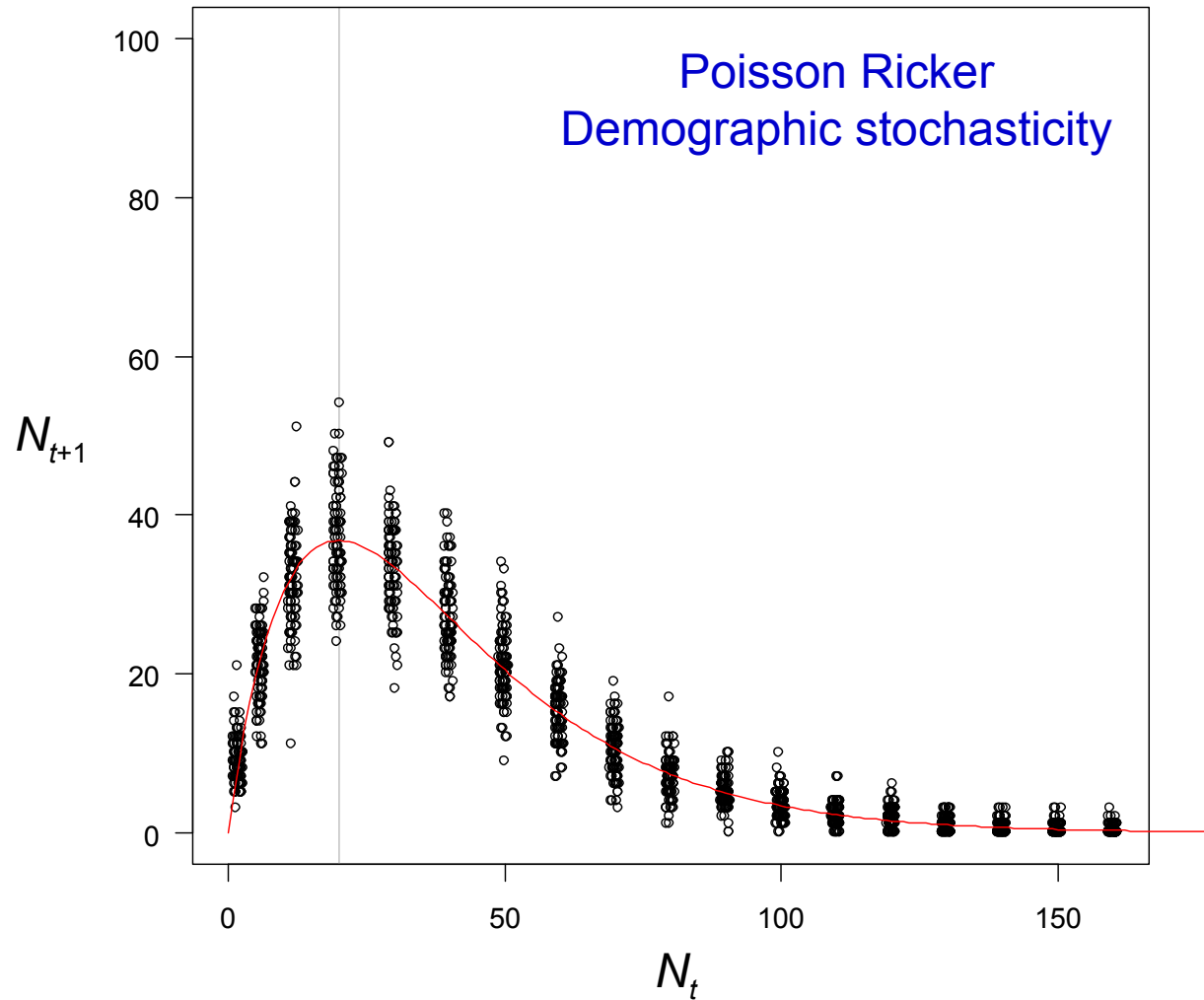
Model for demographic stochasticity



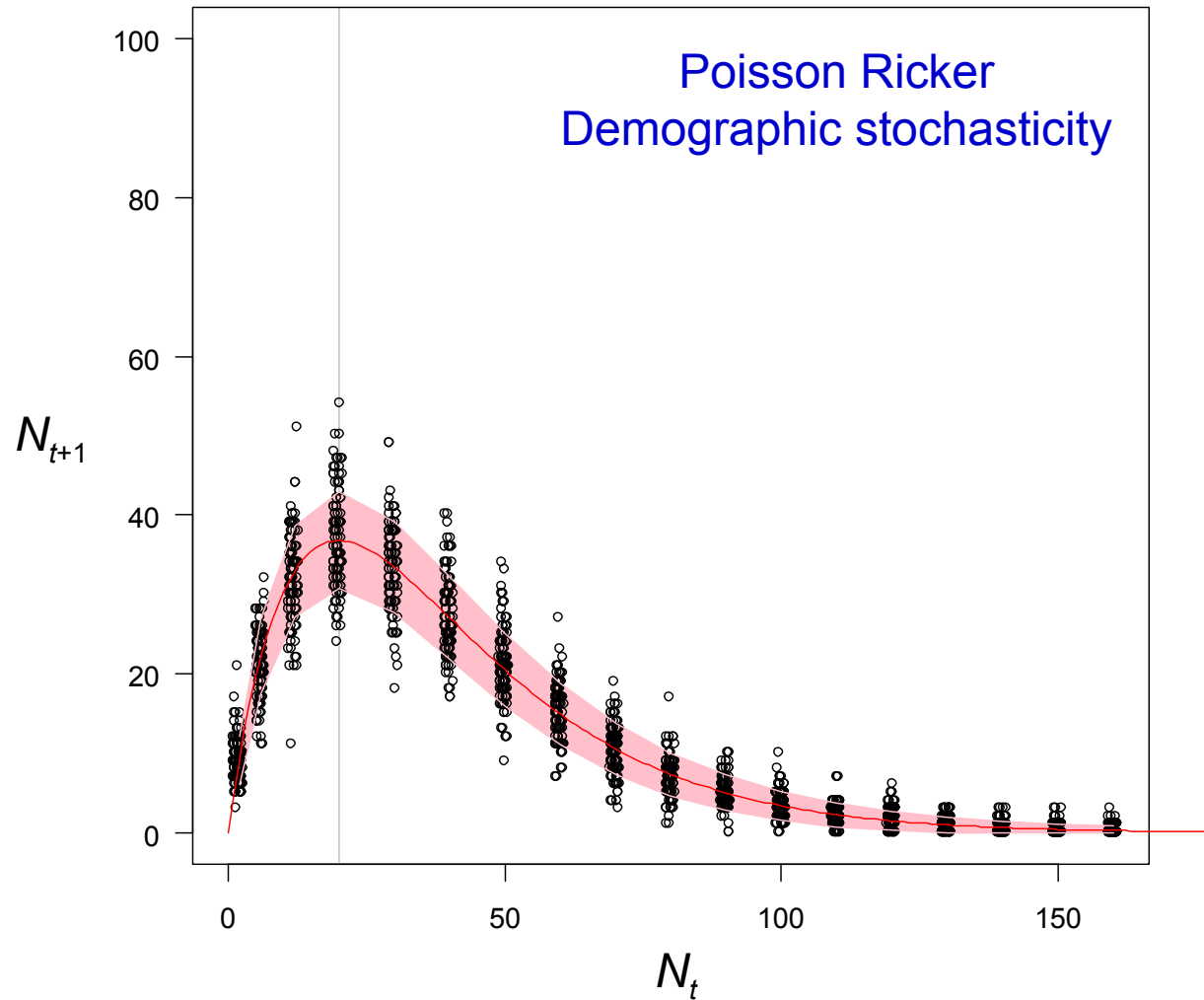
Demographic stochasticity	Environmental stochasticity	Demographic heterogeneity	Sex	Model	Abbr.
●				Poisson	P
●	●			Neg bin-environmental	NBe
●		●		Neg bin-demographic	NBd
●	●	●		Neg bin-gamma	NBg
●			●	Poisson-binomial	PB
●	●		●	Neg bin-binomial-environmental	NBBe
●		●	●	Neg bin-binomial-demographic	NBBd
●	●	●	●	Neg bin-binomial-gamma	NBBg

All models have mean: $N_{t+1} = N_t R e^{-\alpha N_t}$

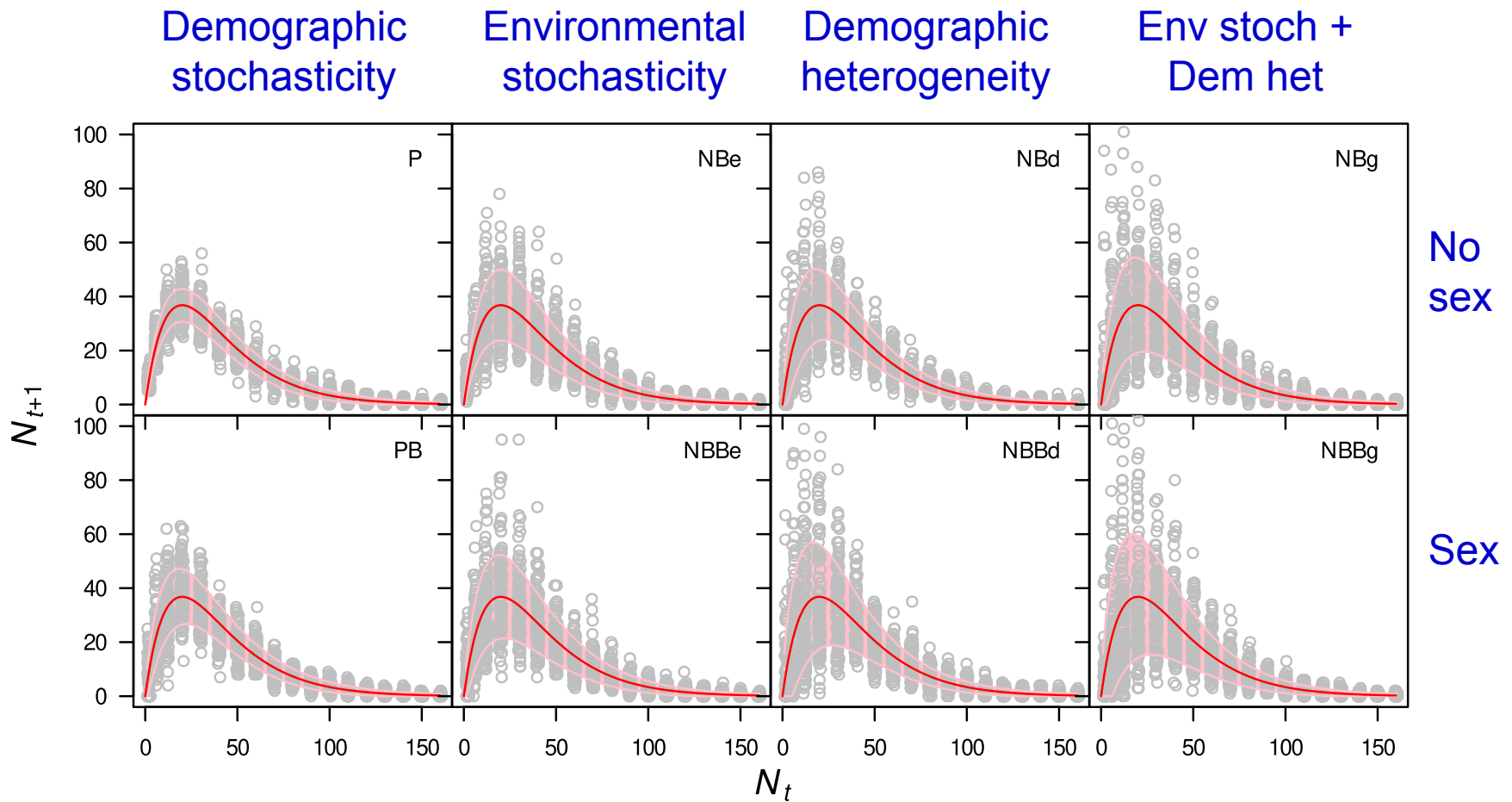
Stochastic production functions



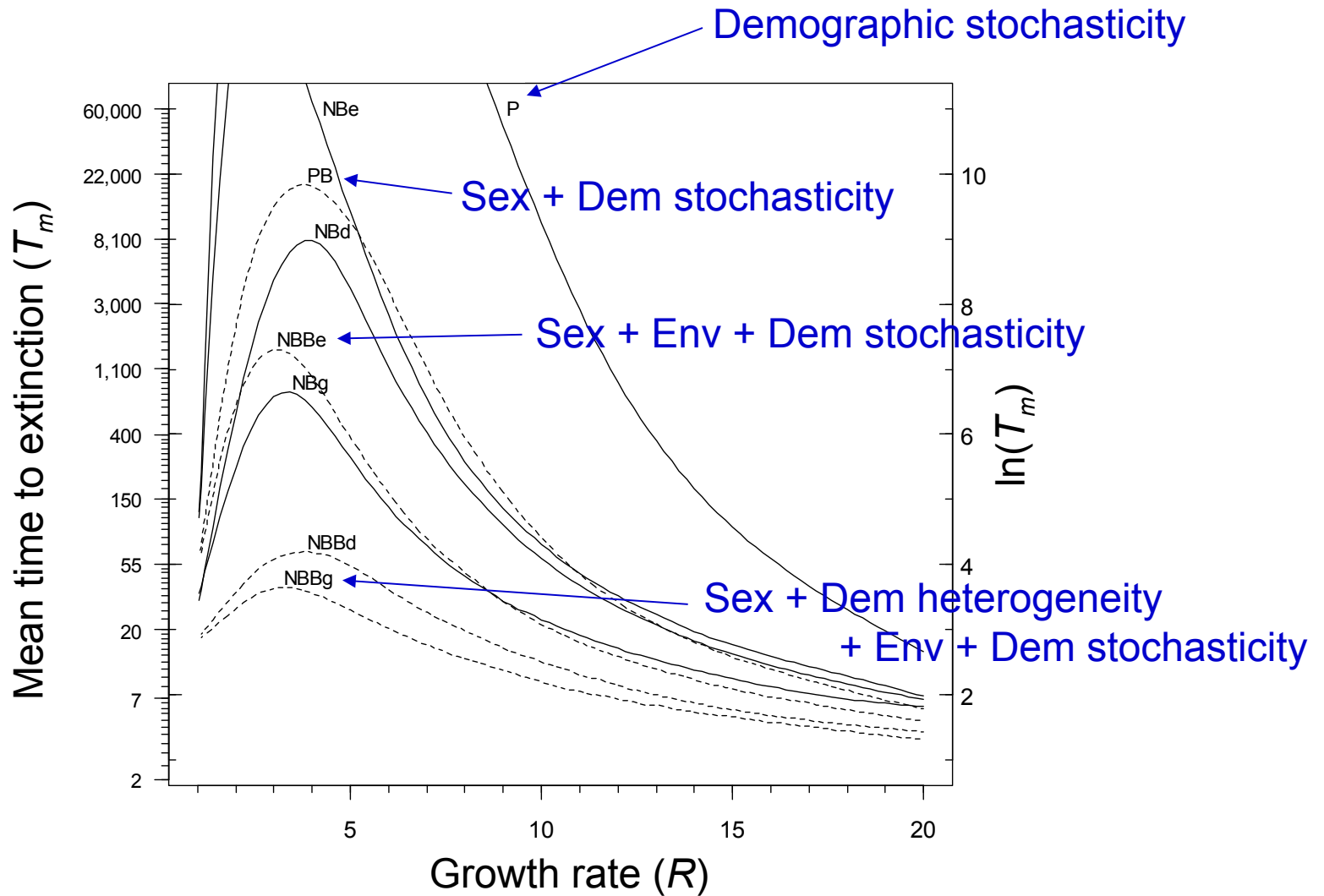
Stochastic production functions



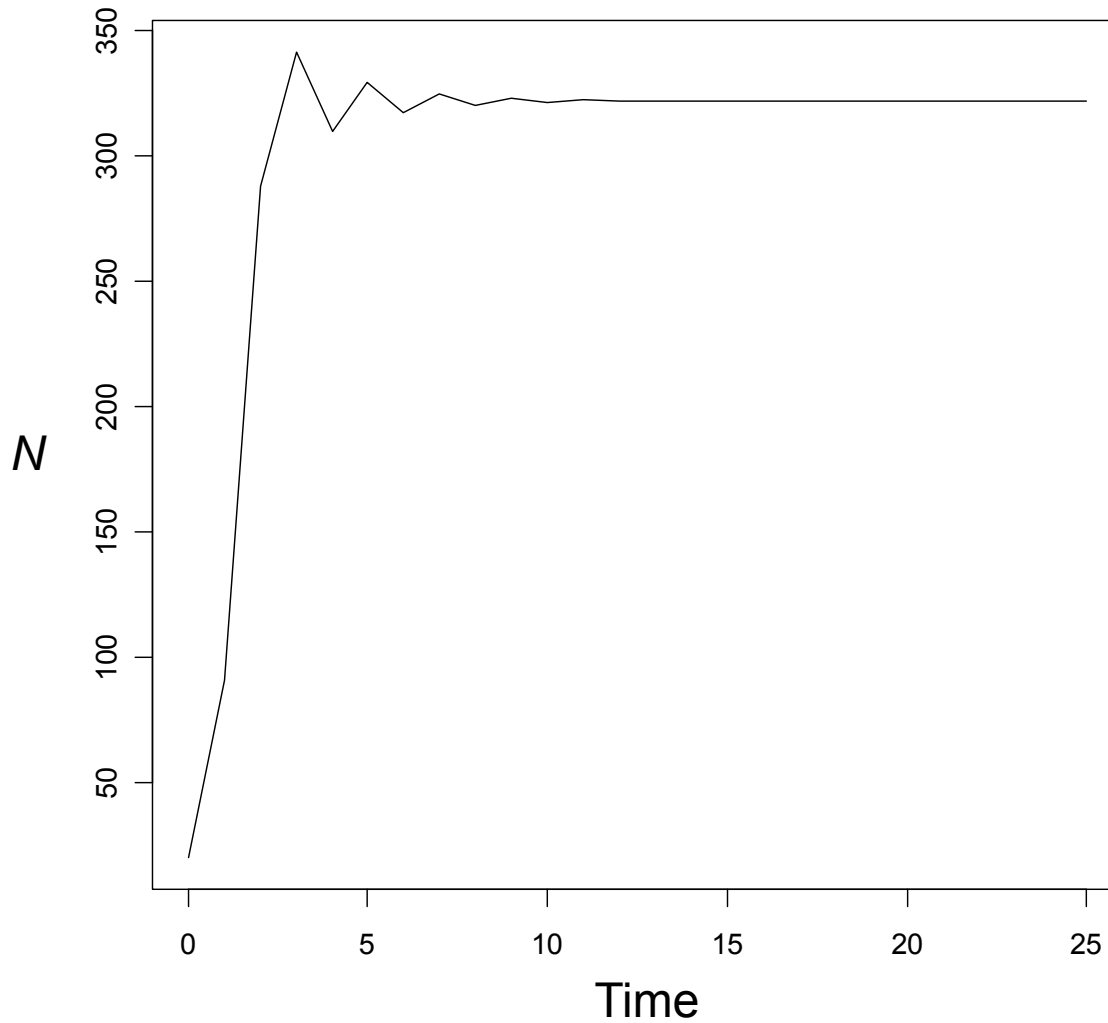
Stochastic production functions



Extinction times



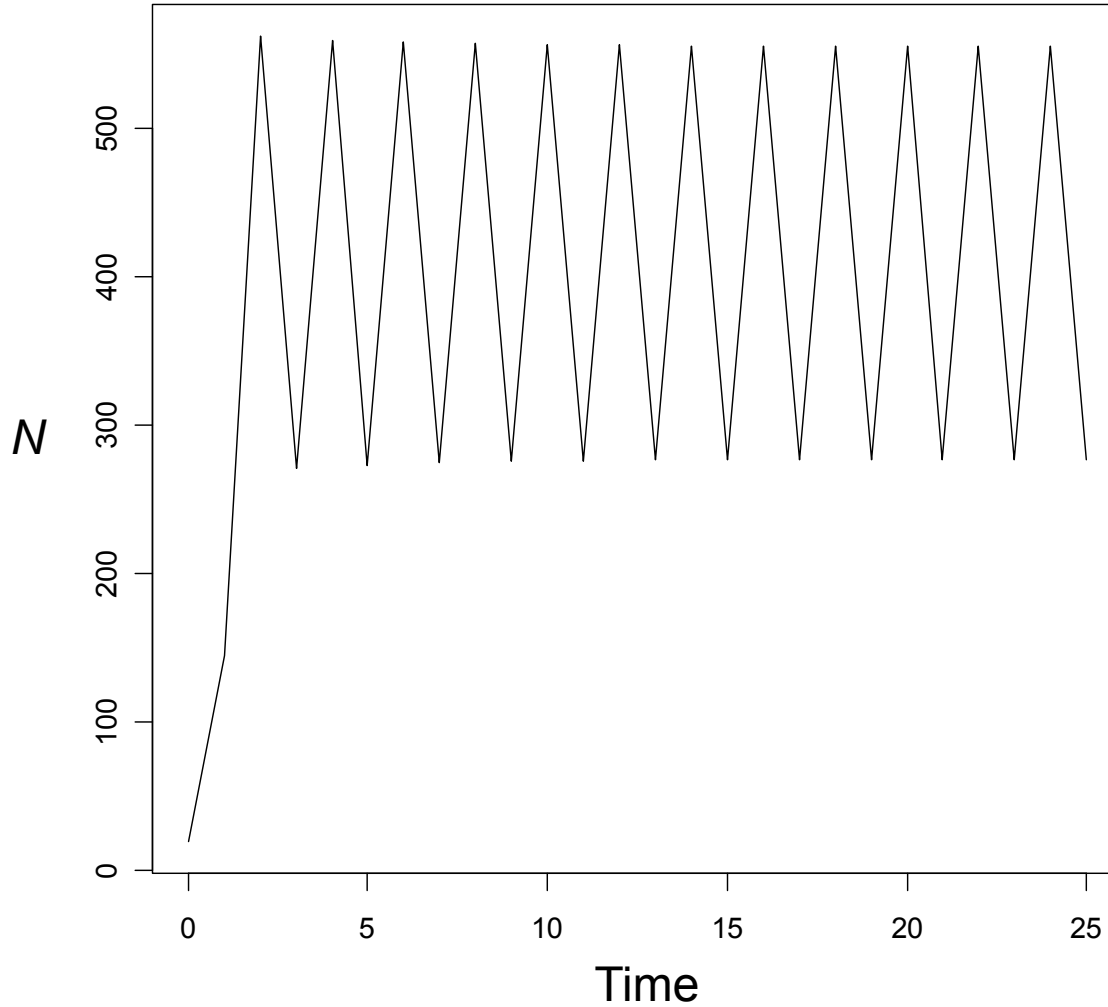
Nonlinear dynamics



$R = 5$

Stable
equilibrium

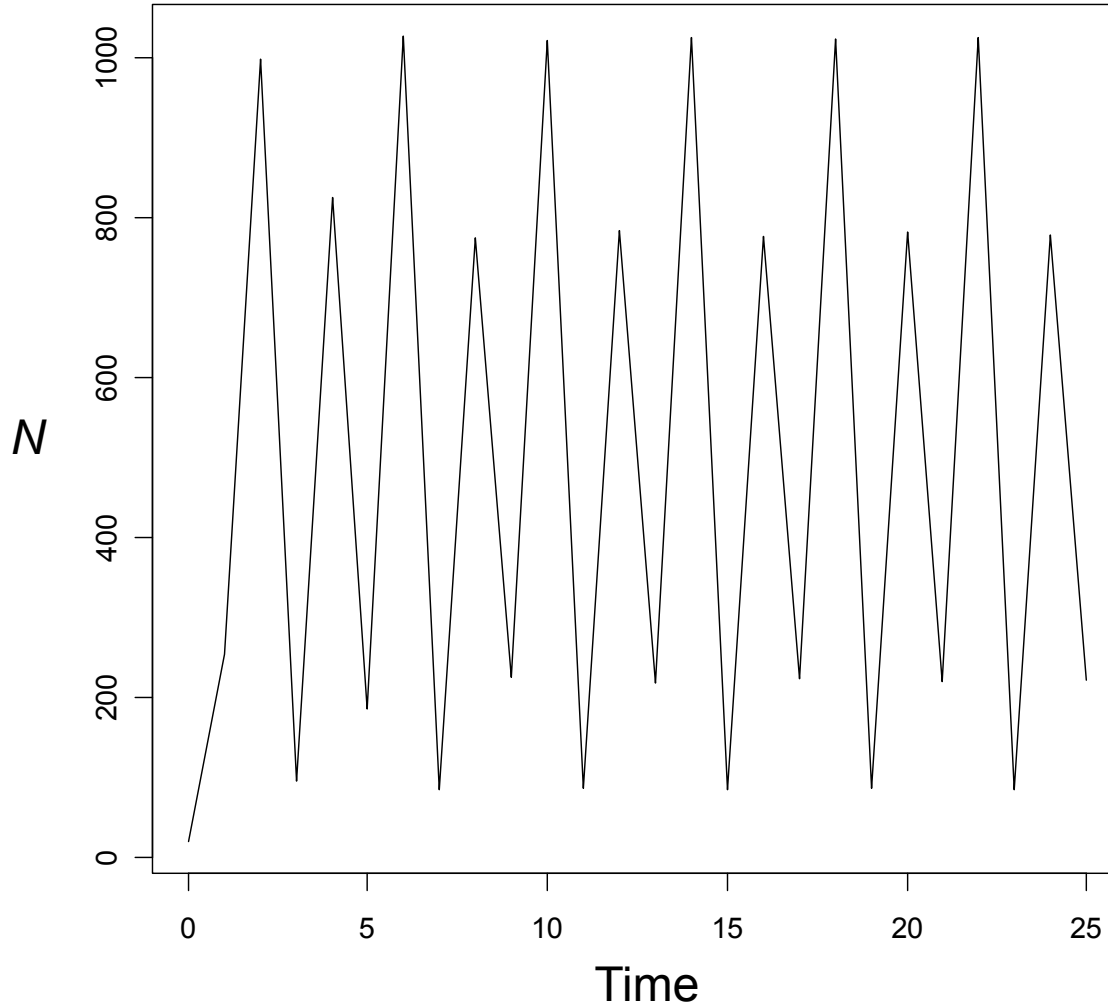
Nonlinear dynamics



$R = 8$

Two point
limit cycle

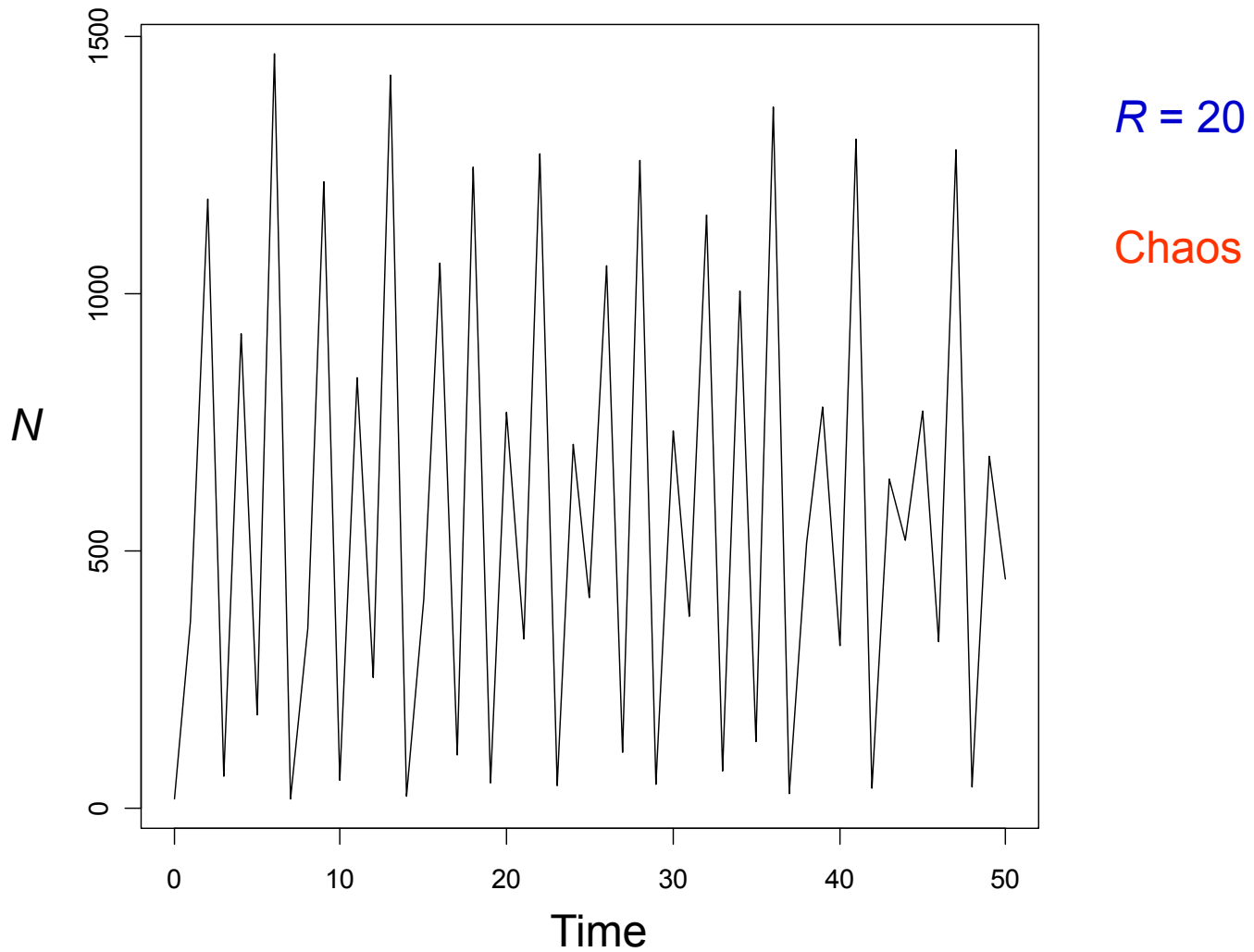
Nonlinear dynamics



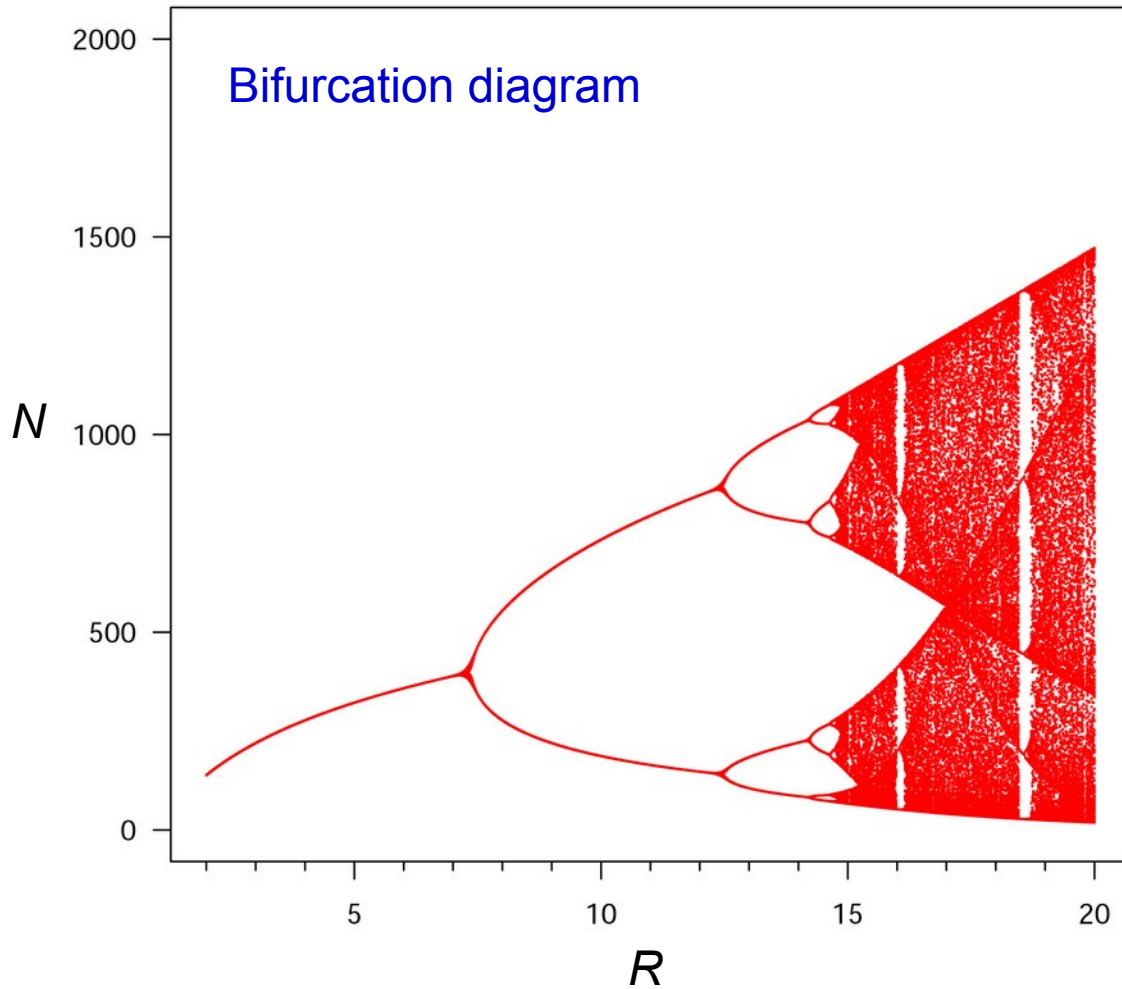
$R = 14$

Four point
limit cycle

Nonlinear dynamics



Nonlinear dynamics



Stochastic nonlinear dynamics

